



Chapter Notes: A Journey through States of Water

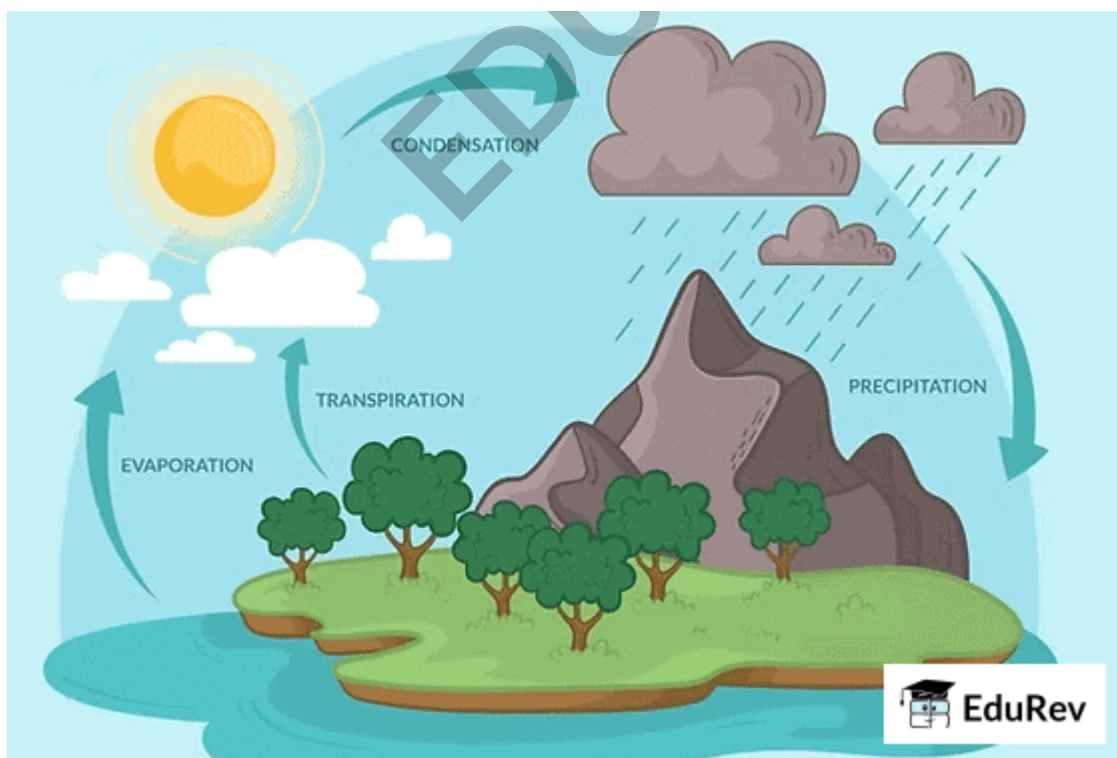
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Introduction

Have you ever wondered what makes up most of **our bodies** and the **Earth's surface**? The answer is **water**. Water is a clear liquid that we drink every day. It covers most of the Earth and is **important** for all living things to stay healthy and alive. Our bodies need water to work properly- for example, to help digest food and to cool the body. Plants, animals and humans all need water to **survive**. Without water, life on Earth would not be possible. Water is also used for washing, cooking and cleaning.



Think about it-every living creature, from the tiniest insect to the largest animal, needs water to stay alive and healthy.

Are Ice and Water the Same Substance?

Ice and water are two different **states** of the same substance. They look and behave differently, but both are made of the same chemical-water. The difference is in how the water molecules are arranged and how much energy (heat) they have.

Liquid water: It flows and takes the shape of its container. Its molecules move more freely and can slide past one another.

Ice (solid water): It holds its shape and does not flow. The molecules are arranged in a fixed, rigid pattern and have less energy.



So, while ice and liquid water behave differently, they are the same substance in different states because the arrangement and energy of the water molecules are different.

MULTIPLE CHOICE QUESTION

Try yourself: What do ice and water represent?

- A Two different substances
- B Two different forms of the same substance
- C Two separate elements
- D Two types of animals

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Investigating Water's Disappearing Act

Where does the water go after the rain stops? Why does a little water sprinkled on a hot pan seem to disappear while making dosa? The water does not vanish-its form changes.

When water on the hot pan is heated, it turns into **steam** or **water vapour**. Water vapour is an invisible gas; when it cools and forms tiny liquid droplets, these become visible as steam.



Evaporation: Water Turning into Vapour

Evaporation is the process by which liquid water becomes water vapour (a gas).

Evaporation happens continuously, even at room temperature; higher temperature increases the rate.

Examples: drying wet clothes, water from puddles disappearing, and sweat evaporating from our skin.



Why Puddles Disappear

Some water soaks into the ground (this is called **seepage** or **infiltration**).

Some water changes into water vapour by **evaporation** and rises into the air.

MULTIPLE CHOICE QUESTION

Try yourself: Which process describes the change of water into a gas called water vapor?

A Condensation

B Evaporation

C Freezing

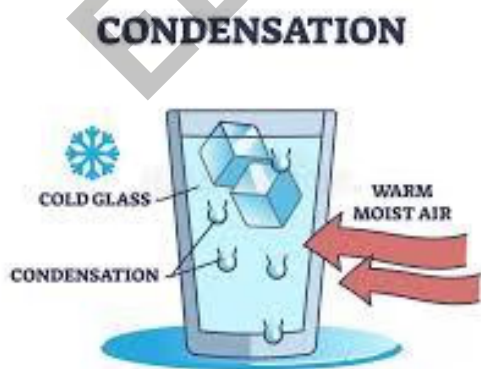
D Melting

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Condensation: Vapour Turning Back into Liquid

Have you noticed droplets forming on the outside of a cold water bottle taken out of the fridge? Those droplets are formed by **condensation**.

Water droplets form on the cold bottle because water vapour in the air touches the cold surface and changes into liquid water. This process is called condensation.



Condensation occurs when warm, moist air cools down and the water vapour in it turns back into tiny liquid droplets.

These tiny droplets collect on cold surfaces such as glass, leaves or metal and become visible.



Humidity is the amount of water vapour present in the air. High humidity means the air already contains a lot of water vapour; low humidity means it contains little.

What Are the Different States of Water?



Water exists in three common states that we see in daily life: **solid**, **liquid** and **gas**.

Solid (ice): Keeps its shape and does not flow. Examples of solids in daily life include ice, stones, wood and glass.

Liquid (water): Takes the shape of its container and flows. Liquids have a fixed volume but change shape to fit their container. Examples: water, milk, oil.

Gas (water vapour): Spreads out to fill the available space and has neither fixed shape nor fixed volume. Water vapour is present in the air even at room temperature but is invisible unless it condenses into droplets (as in steam).

MULTIPLE CHOICE QUESTION

Try yourself: Where does water go when it is sprinkled on a hot pan?

- A It turns into ice.
- B It becomes steam.
- C It disappears completely.
- D It turns into snow

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How Can We Change the State of Water?

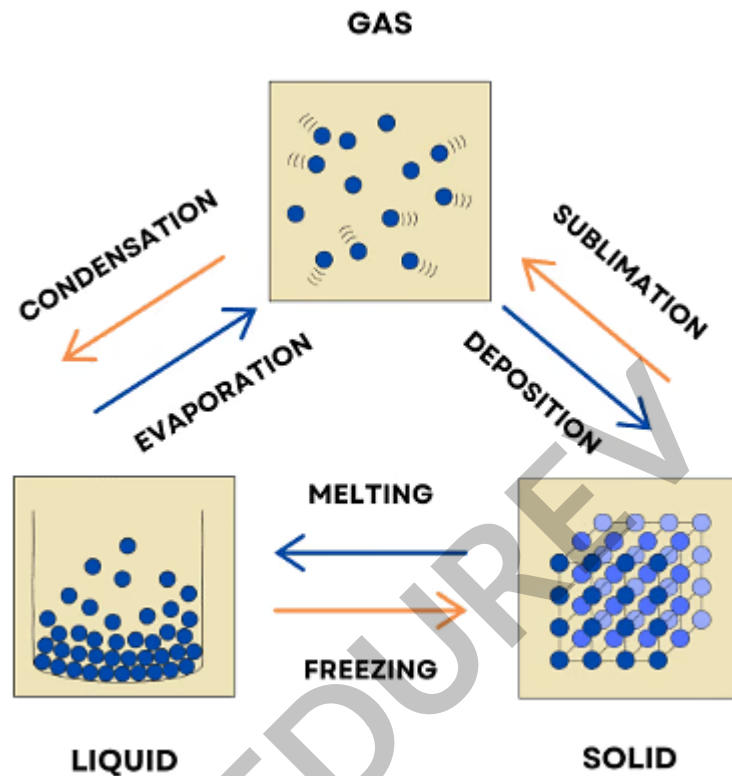
Yes. Adding or removing heat changes the state of water. The common state changes are:

Melting: Changing from solid (ice) to liquid (water) by adding heat. The melting point of pure water at 1 atmosphere pressure is **0°C**.

Freezing: Changing from liquid (water) to solid (ice) by removing heat (cooling).

Evaporation / Boiling: Changing from liquid (water) to gas (water vapour) by adding heat. Boiling occurs when bubbles of vapour form within the liquid; the boiling point of pure water at 1 atmosphere pressure is **100°C**.

Condensation: Changing from gas (water vapour) back to liquid by cooling.



MULTIPLE CHOICE QUESTION

Try yourself: What is the process of water changing from a gaseous state to a liquid state?

A Melting

B Freezing

C Evaporation

D Condensation

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How Can Evaporation Be Faster or Slower?

Why do clothes dry faster on a windy day?



Clothes dry faster on a windy day because moving air removes the moisture from the cloth and replaces it with drier air, increasing the rate of evaporation.



Factors affecting evaporation:

Temperature: Higher temperature increases the rate of evaporation because particles gain more energy and escape as vapour.

Air movement (wind): Faster air movement carries away saturated air and replaces it with drier air, increasing evaporation.

Surface area: A larger surface area allows more molecules to escape, so water evaporates quicker.

Humidity: High humidity (more water vapour already in the air) reduces evaporation; low humidity increases it.

Humidity on rainy days

When the air already contains a lot of water vapour (high humidity), evaporation slows down.

This is why clothes and sweat take longer to dry on rainy or humid days.

Cooling Effect of Evaporation

Why do we feel cooler under a fan? Moving air increases evaporation of sweat from our skin. Evaporation requires heat energy; this heat is taken from the skin and surroundings, so we feel cooler.



Examples of cooling by evaporation:

Sweating: Evaporation of sweat from the skin removes heat and cools the body.

Earthen pots: Water seeping through the porous surface of an earthen pot evaporates, keeping the stored water cool.

Sprinkling water: Sprinkling water on a floor or roof in hot weather cools the area as the water evaporates.

MULTIPLE CHOICE QUESTION

Try yourself: How does water change from solid to liquid?

A By heating

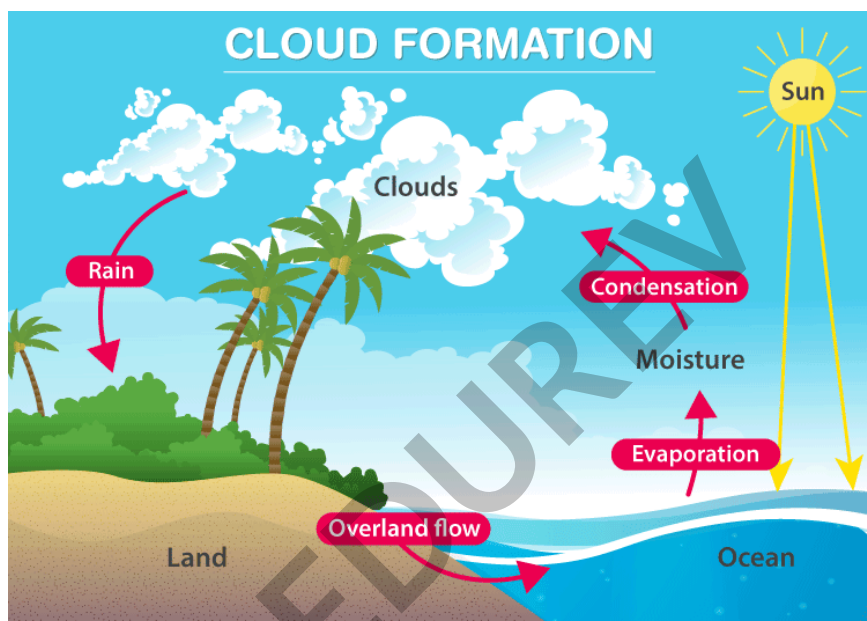
B By cooling

C By freezing

D By evaporating

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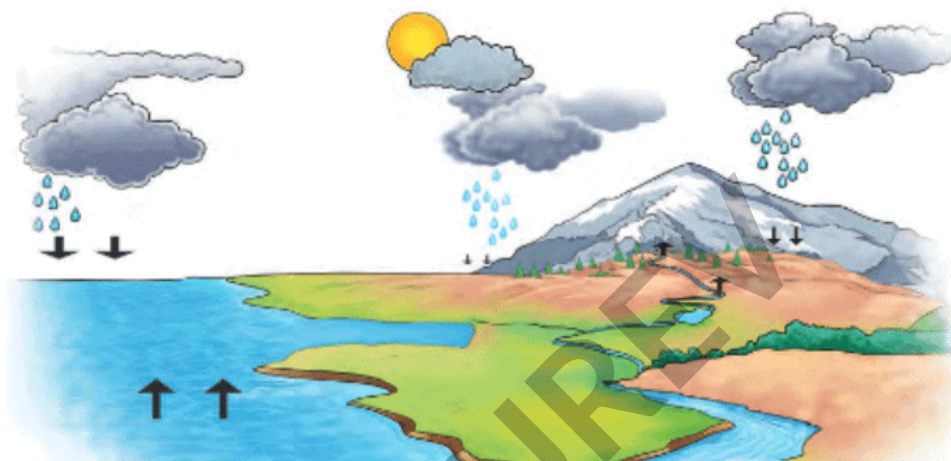
How Do Clouds Give Us Rain?



Role of Condensation in the Water Cycle

Evaporation sends water vapour into the atmosphere from oceans, lakes, soil and plants. As moist air rises, it cools. When this air cools enough, water vapour condenses into tiny liquid droplets around dust or smoke particles (these particles are called condensation nuclei).

These tiny droplets join together to form clouds. When they grow large and heavy enough, they fall as rain (precipitation).



Water Availability and Conservation

Only a small portion of the Earth's water is fresh and available for use by **plants, animals and humans**; most water is in the oceans and is salty.

Water is essential for drinking, farming and many daily activities. Growing demand and pollution lead to water shortages in many places.

It is important to use water wisely, avoid wastage and keep rivers, lakes and groundwater free from pollution.

Why does water vapour rise?

Air that contains a lot of **water vapour** can become **lighter** (less dense) than surrounding air. This lower density allows the moist air to rise higher into the atmosphere, similar to how a lighter gas makes a balloon rise.



MULTIPLE CHOICE QUESTION

Try yourself: Which of the following factors will increase the rate of evaporation of water?

- A Lower temperature
- B Larger surface area
- C Higher humidity
- D Decreased airflow

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Key Points

Evaporation: Liquid water changes into water vapour.

Condensation: Water vapour changes back into liquid water.

States of water: Solid (ice), liquid (water), gas (water vapour).

State changes: Melting (solid $\rightarrow$ liquid), freezing (liquid $\rightarrow$ solid), evaporation (liquid $\rightarrow$ gas), condensation (gas $\rightarrow$ liquid).

Factors affecting evaporation: Temperature, surface area, air movement and humidity.

Cooling effect: Evaporation absorbs heat from the surroundings and causes cooling.

Cloud and rain formation: Caused by condensation of water vapour and the joining of droplets.

Water cycle: Continuous movement of water between Earth's surface and the atmosphere through evaporation, condensation and precipitation.

You can practice questions from this chapter here: [Very Short Question Answers: A Journey through States of Water](https://edurev.in/studytube/edurev/32ba9d1c-ae0b-454d-9f3f-7aed1b1313e7_t) (https://edurev.in/studytube/edurev/32ba9d1c-ae0b-454d-9f3f-7aed1b1313e7_t)